

2.10

$$\begin{cases} -2x - 4y + 6z = 4 \\ -3x - 3y + 8z = 4 \\ 3y - z = k \end{cases}$$

para que valores de k el sistema es compatible determinado (única solución)

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$$\begin{pmatrix} -2 & -4 & 6 \\ -3 & -3 & 8 \\ 0 & 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ k \end{pmatrix}$$

$$A\vec{x} = \vec{b} \Rightarrow \underbrace{A^{-1}A}_{I}\vec{x} = A^{-1}\vec{b} \Rightarrow \vec{x} = A^{-1}\vec{b}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{Adj}(A)$$

$$\det(A) = \begin{vmatrix} -2 & -4 & 6 \\ -3 & -3 & 8 \\ 0 & 3 & -1 \end{vmatrix} = -6 - 54 + 0 - (0 - 48 - 12) = 0$$

No es un sistema compatible
(incompatible o compatible indeterminado)

b) Para que valores de k el sistema es compatible indeterminado

$$\begin{pmatrix} -2 & -4 & 6 \\ -3 & -3 & 8 \\ 0 & 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ k \end{pmatrix}$$

$$\begin{cases} -2x - 4y + 6z = 4 & (1) \\ -3x - 3y + 8z = 4 & (2) \\ 0x + 3y - z = k & (3) \end{cases} \Rightarrow$$

$$\begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix}^{\vec{n}_1} x + \begin{pmatrix} -4 \\ -3 \\ 3 \end{pmatrix}^{\vec{n}_2} y + \begin{pmatrix} 6 \\ 8 \\ -1 \end{pmatrix}^{\vec{n}_3} z = \begin{pmatrix} 4 \\ 4 \\ k \end{pmatrix}$$

$$x \vec{n}_1 + y \vec{n}_2 + z \vec{n}_3 = \vec{b}$$

$$\text{de (3)} \quad z = 3y - k \Rightarrow \begin{cases} -2x - 4y + 6(3y - k) = 4 \\ -3x - 3y + 8(3y - k) = 4 \end{cases}$$

$$\begin{cases} -2x + 14y = 4 + 6K \\ -3x + 21y = 4 + 8K \end{cases} \Rightarrow \begin{cases} -x + 7y = 2 + 3K \\ -x + 7y = \frac{4 + 8K}{3} \end{cases}$$

$$2 + 3K = \frac{4 + 8K}{3} \Rightarrow 6 + 9K = 4 + 8K$$

$$\boxed{K = -2}$$

Verificar $K = -2$

$$\begin{cases} -2x - 4y + 6z = 4 \\ -3x - 3y + 8z = 4 \\ 3y - z = -2 \end{cases}$$

$$\Rightarrow \begin{cases} x - y - 2z = 0 \\ 3y - z = -2 \Rightarrow \boxed{z = 3y + 2} \end{cases}$$

$$x - y - 2(3y + 2) = 0$$

$$x - 7y - 6 + 4 = 0$$

$$\boxed{x = 2 + 7y}$$

$$\boxed{\vec{x} = \begin{pmatrix} 2 + 7y \\ y \\ 3y + 2 \end{pmatrix}}$$

$$\Rightarrow \vec{x} = \begin{pmatrix} 2y \\ y \\ 3y \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

Por $K \neq -2$ el sistema es incompatible

$$\Rightarrow AX = \vec{0}$$

$$\begin{cases} -2x - 4y + 6z = 0 \\ -3x - 3y + 8z = 0 \\ 3y - z = 0 \end{cases}$$

$$\Rightarrow x - y - 2z = 0 \Rightarrow x - y - 6y = 0$$

$$x = 7y$$

$$3y - z = 0 \Rightarrow z = 3y$$

$$\text{Sol: } \vec{x}_h = \begin{pmatrix} 7y \\ y \\ 3y \end{pmatrix}$$

guia 3

1) $V = \{ f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ é função} \}$

$$+ : (f+g)(x) = f(x) + g(x) \quad \forall x \in \mathbb{R}$$
$$\cdot : (k \cdot f)(x) = k f(x) \quad \forall x \in \mathbb{R}$$

seja $f, g, h \in V$; $0, b \in \mathbb{R}$

1) $(f + (g+h))(x) = f(x) + (g+h)(x) = f(x) + g(x) + h(x)$
 $= (f+g)(x) + h(x) = ((f+g) + h)(x) \quad (\text{associativity})$

2) $(f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x) \quad (\text{commutativity})$

3) seja $z(x) = 0 \quad \forall x \in \mathbb{R} \quad \therefore z(x) \in V$

$$(z+f)(x) = z(x) + f(x) = 0 + f(x) = f(x)$$

4) seja $z(x) = -f(x)$

$$(f+z)(x) = f(x) + z(x) = f(x) + (-f(x)) = f(x) - f(x) = 0$$

5) $(a(bf))(x) = a \cdot (bf)(x) = a b f(x) =$
 $= (ab)f(x)$

6) $(1f)(x) = 1 f(x) = f(x)$

7) $(a(f+g))(x) = a(f+g)(x) = a(f(x) + g(x)) =$
 $a f(x) + a g(x) = (af + ag)(x)$

$$8) ((a+b)f)(u) = (a+b)f(u) = af(u) + bf(u) \\ = (af + bf)(u)$$

$\therefore (V; +; R; \cdot)$ es espacio vectorial

$$4.2) A = \{ (x_1, x_2) \in \mathbb{R}^2 / x_1 = 0 \}$$

Ver si A es un subespacio de $\mathbb{R}^2$

Diremos que $(S \subseteq V; +; K; \cdot)$ es un subespacio vectorial sobre un cuerpo de escalares K

$$\bullet S \neq \emptyset$$

$$v_1, v_2 \in S \Rightarrow (v_1 + v_2) \in S$$

$$\lambda \in K \quad v \in S \Rightarrow (\lambda v) \in S$$

$$\text{--- o --- o ---}$$

$$\text{Ej } v = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in S \quad \therefore S \neq \emptyset$$

$$\text{Como son todos los vectores de } S \Rightarrow v = \begin{pmatrix} 0 \\ \alpha \end{pmatrix} \text{ con } \alpha \in \mathbb{R}$$

$$\therefore \vec{0} \in S$$

$$v_1, v_2 \in S \Rightarrow v_1 = \begin{pmatrix} 0 \\ \alpha_1 \end{pmatrix} \quad v_2 = \begin{pmatrix} 0 \\ \alpha_2 \end{pmatrix} \text{ con } \alpha_1, \alpha_2 \in \mathbb{R}$$

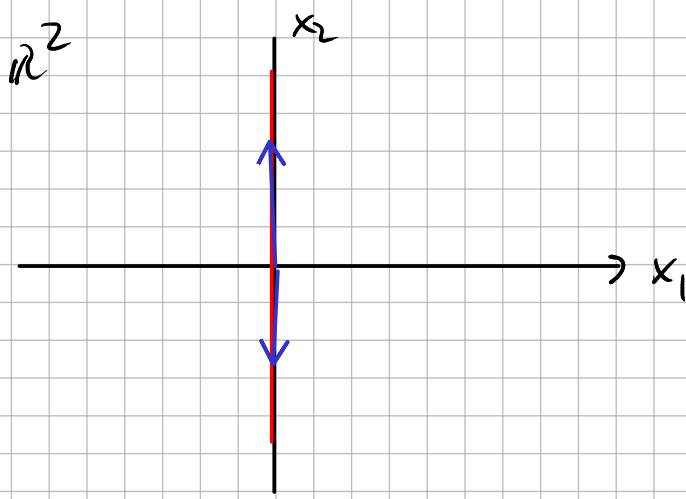
$$v_3 = (v_1 + v_2) = \begin{pmatrix} 0 \\ \alpha_1 \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \alpha_1 + \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \alpha_3 \end{pmatrix} \text{ con } \alpha_3 \in \mathbb{R}$$

$\underbrace{\alpha_3}_{\in S}$

$$v_4 = (\lambda v_1) = \lambda \begin{pmatrix} 0 \\ \alpha_1 \end{pmatrix} = \begin{pmatrix} 0 \\ \lambda \alpha_1 \end{pmatrix} = \begin{pmatrix} 0 \\ \alpha_4 \end{pmatrix} \text{ con } \alpha_4 \in \mathbb{R}$$

$\underbrace{\alpha_4}_{\in S}$

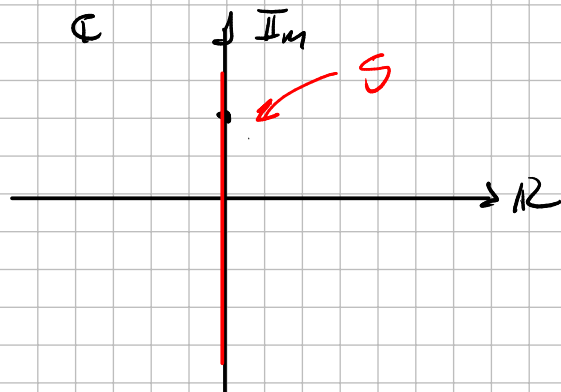
$\therefore S$ es un subespacio vectorial



5) d) $V = \mathbb{C}$ $K = \mathbb{R}$ $S = \{ ai / a \in \mathbb{R} \}$

Como son los elementos de S

son números complejos $z = ai$



$z_1, z_2 \in S$ y $\alpha \in \mathbb{R}$

con $z_1 = x_1 i$ $z_2 = x_2 i$ $| S \neq \emptyset$

$z_4 = (z_1 + z_2) = x_1 i + x_2 i = \underbrace{(x_1 + x_2)}_{\in \mathbb{R}} i$

$\Rightarrow \boxed{z_4 \in S}$ ✓

$z_5 = (\lambda z_1) = \lambda z_1 = \lambda x_1 i = \underbrace{(\lambda x_1)}_{\in \mathbb{R}} i$

$\boxed{z_5 \in S}$ ✓

$\Rightarrow S$ es sub espacio vectorial de $\mathbb{C}$

e) $V = \mathbb{C}$ $K = \mathbb{C}$ $S = \{ ai \text{ con } a \in \mathbb{R} \}$

sea z_1 y $z_2 \in S$ y $\alpha \in \mathbb{C}$

$z_1 = x_1 i$ $z_2 = x_2 i$ con $x_1, x_2 \in \mathbb{R}$

$$z_3 = (z_1 + z_2) = z_1 + z_2 = x_1 i + x_2 i = \underbrace{(x_1 + x_2)}_{\in \mathbb{R}} i$$

$$\Rightarrow z_3 \in S$$

$$z_4 = (\alpha z_1) = \alpha z_1 = (a + ib) x_1 i = (a x_1) i - b x_1$$

$$z_4 \notin S$$

$\Rightarrow S$ no es un subespacio vectorial de $(\mathbb{C}; +; \cdot)$

$$\text{ej) } \left\{ \underbrace{\begin{pmatrix} 1 \\ 2 \\ \kappa \end{pmatrix}}_{\vec{n}_1} \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{\vec{n}_2} \begin{pmatrix} 0 \\ 1 \\ 1-\kappa \end{pmatrix} \right\}$$

$$c_1 \vec{n}_1 + c_2 \vec{n}_2 + \dots + c_n \vec{n}_n = \vec{0}$$

$$\text{so } \{\vec{n}_1; \vec{n}_2; \dots; \vec{n}_n\} \text{ son LI} \Rightarrow c_1 = c_2 = c_3 = \dots = c_n = 0$$

$$\alpha \vec{n}_1 + \beta \vec{n}_2 + \gamma \vec{n}_3 = \vec{0}$$

$$\alpha \begin{pmatrix} 1 \\ 2 \\ \kappa \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 1 \\ 1-\kappa \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} \alpha + \beta + 0\gamma = 0 \\ 2\alpha + \beta + \gamma = 0 \\ \kappa\alpha + \beta + \gamma(1-\kappa) = 0 \end{cases}$$

$$\Rightarrow \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ \kappa & 1 & 1-\kappa \end{pmatrix}}_A \underbrace{\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}_{\vec{x}} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}}_{\vec{0}}$$

$$A \vec{x} = \vec{0} \quad \text{systema homogéneo}$$

so el $\det(A) \neq 0 \rightarrow A$ es invertible $\rightarrow \vec{x} = \vec{0}$

$\rightarrow \alpha=0 ; \beta=0 \gamma=0 \rightarrow N_1; N_2; N_3$ son LI

$$\det(A) = \begin{vmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ \kappa & 1 & 1-\kappa \end{vmatrix} = ((1-\kappa) + 0 + \kappa) - (1 + 2(1-\kappa))$$

$$= \cancel{1-\kappa} + \cancel{\kappa} - 1 - 2 + 2\kappa = -2 + 2\kappa = 2(\kappa - 1)$$

$\therefore$ cuando $\kappa \neq 1 \rightarrow \det(A) \neq 0 \rightarrow \begin{cases} \alpha=0 \\ \beta=0 \\ \gamma=0 \end{cases} \Rightarrow$

$N_1; N_2; N_3$ son LI

con $\kappa=1$ no son LI (son LD)

0
tomo $\kappa=1$ (verificación)

$$N_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \quad N_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad N_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$N_3 = N_1 - N_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$